

Identifying in-patient costs attributable to the clinical sequelae and comorbidities of alcoholic liver disease in a national hospital database

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ABSTRACT

Background and Aims The clinical sequelae and comorbidities of alcoholic liver disease (ALD) often require hospitalization. The aims of this study were to (1) compare the average costs of hospitalizations with ALD and the costs of hospitalizations with other alcohol-related diagnoses that do not involve the liver; and (2) estimate the percentage of the difference in costs between the ALD and non-ALD hospitalizations that may be attributed to ascites, protein-calorie malnutrition and other conditions. **Design** The 2012 National Inpatient Sample is a population-based cross-sectional database representing more than 94% of all discharges from community hospitals in the United States. **Setting** Community hospitals in the United States. **Participants** The sample included 72 531 hospitalizations with ALD and 287 047 hospitalizations with other alcohol-related diagnoses. **Measurements** The dependent variable was total in-patient costs. We estimated the contribution of ascites, protein-calorie malnutrition and other conditions to the difference in costs between patients with ALD and patients with other diagnoses. **Findings** Average costs for ALD patients were \$3188.4 higher than those for patients with other diagnoses (\$13 543 versus \$10 355; $P < 0.001$). Among all conditions in the analysis, protein-calorie malnutrition had the largest impact on costs [\$6501; 95% confidence interval (CI) = 5956, 7045; $P < 0.001$] accounting for 12% of the higher costs of ALD stays. **Conclusions** Costs of hospital care for patients with alcoholic liver disease are higher than those for patients with other alcohol-related diagnoses. These increased costs are associated with specific clinical sequelae and comorbidities, with protein-calorie malnutrition—a largely preventable condition—making a substantial contribution.

Keywords Alcoholic liver disease, comorbidity, costs, decomposition, hospital, malnutrition.

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INTRODUCTION

Globally, the burden of alcoholic cirrhosis and other forms of alcoholic liver disease (ALD) is substantial, resulting in an estimated 494 000 deaths in 2010 [1]. In the United States, the increasing consumption of alcohol [2,3] suggests that the costs of treating ALD will persist well into the future. Between 2002 and 2010, the number of hospitalizations involving alcoholic hepatitis in the United States increased by 30% [4]. Cirrhosis of the liver was the eighth leading cause of years of life lost in 2010 [5], and approximately 60% of cirrhosis hospitalizations include at least one alcohol-related diagnosis [6]. Further, the changing demographic and clinical profile of liver disease patients has implications for hospital costs. The annual number of

hospitalizations involving cirrhosis increased by 35% between 2002 and 2010, from approximately 170 400 to 230 800 discharges [7]. During this same time-period, the population hospitalized with this condition became older and more severely ill; specifically, the percentage of hospitalized cirrhosis patients aged more than 50 years increased by 10%, and comorbid conditions in this population increased by 40% [6]. These trends underscore some of the challenging quality and resource management issues that face hospitals caring for individuals with alcohol use disorders [8].

Alcoholic liver disease encompasses three main types of liver injury: alcoholic fatty liver, alcoholic hepatitis and alcoholic cirrhosis [9]. Alcoholic fatty liver refers to the accumulation of fats in the liver due to alcohol-induced

alterations in metabolism. With continued alcohol misuse patients with fatty liver may develop alcoholic hepatitis, which is characterized by inflammation and some degree of fibrosis, a wound-healing response to tissue damage [10]. In alcoholic cirrhosis, healthy tissue is replaced with scarring that impedes the flow of blood through the liver and slows the processing of nutrients, hormones, drugs and toxins. Fatty liver and alcoholic hepatitis may be reversible with abstinence, whereas cirrhosis is irreversible [11,12].

The sequelae of ALD often require hospitalization. Patients with alcoholic hepatitis or cirrhosis may develop ascites, a build-up of fluid in the abdomen that increases mortality risk [11]. Impaired liver functioning can also cause hepatic encephalopathy, a brain disorder that in its advanced stages leads to hepatic coma or loss of consciousness. Cirrhosis is a risk factor for hepatocellular carcinoma—the most common type of liver cancer—as well as portal hypertension, a condition in which pressure increases in the vein that carries blood from the digestive organs to the liver. To compensate for the blockage of blood flow through the liver, the esophagus and stomach may develop veins that become fragile and bleed easily, making upper gastrointestinal (GI) bleeding another serious consequence of ALD.

The extent of liver injury among heavy drinkers varies with comorbid medical conditions. Alcohol abuse contributes to malnutrition generally by suppressing appetite and interfering with the absorption of nutrients, and mortality increases to almost 80% among ALD patients with severe malnutrition [13,14]. Viral hepatitis is common among people with alcohol use disorders, accelerating the progression of liver fibrosis, cirrhosis and cancer [12]. Acute kidney disease is a significant diagnosis associated with ALD and is associated strongly with mortality in cirrhosis [9].

Alcoholic liver disease is a progressive condition that can, in many cases, be reversed, and population health management strategies both to prevent the development and arrest the progression of ALD could reduce substantially the financial burden of alcohol use disorders on hospitals [8]. Although previous work has compared in-patient costs of patients with ALD and patients with other alcohol-related diagnoses [15], little is known about the degree to which the relatively higher costs of ALD stays are driven by specific clinical sequelae and comorbidities.

The current study has the following two aims: (1) to compare the average costs of hospitalizations involving ALD with those of hospitalizations involving all other alcohol-related conditions; and (2) to estimate the percentage of the difference in costs between the ALD and non-ALD hospitalizations that could be associated with ascites, protein-calorie malnutrition and other conditions. Other alcohol-related diagnoses that do not involve the liver are a relevant comparison group for ALD, because all these

conditions share a key modifiable risk factor: the abuse of alcohol. These two groups of patients come from the same target population of individuals with alcohol use disorders, and the clinical management of these patients often has the same therapeutic goal: a reduction or elimination of harmful alcohol consumption.

METHODS

Sample

The National Inpatient Sample (NIS) is a database of hospitalizations derived from billing data submitted by hospitals to state-wide data organizations. The NIS is a 20% stratified random sample drawn from a sampling frame that in 2012 consisted of 94% of all discharges from community hospitals, defined as non-Federal, short-term, general and other specialty hospitals (excluding hospital units of institutions). All available hospitals were included in the sampling frame. These 4378 hospitals (and all their associated discharges) were categorized first into strata defined by hospital census division, rural/urban location, bed size, teaching status and ownership. Within each of the strata, hospitals (clusters) were then selected at a rate of 100% and discharges within hospitals were sampled randomly at a rate of 20% [16]. The current study is based on the 2012 NIS, which included data on 7 296 968 unweighted discharges. The NIS is consistent with the definition of 'limited data sets' under the US Health Insurance Portability and Accountability Act Privacy Rule and contains no direct patient identifiers.

The analysis was restricted to discharges for non-maternal stays of patients aged 18 years and older with any principal or secondary alcohol-related diagnosis, identified with codes from the International Classification of Diseases, 9th revision, Clinical Modification (ICD-9-CM). The analytical sample size was 359 578 (4.9% of the NIS sample). The NIS data set includes a discharge weight for each sampled hospitalization, which is the estimated number of hospitalizations in the population that each sampled hospitalization represents. Applying discharge weights to the data [17], the sampled discharges represent approximately 1.8 million hospitalizations involving any principal or secondary alcohol-related diagnoses in 2012.

MEASURES

Dependent variable

The NIS contains data on total charges for each discharge. Previous analyses of NIS data on in-patient care for liver disease have often relied on charges [18,19]; one such study of trends in alcoholic hepatitis hospitalizations used the terms 'charges' and 'costs' interchangeably [4]. However, charges represent only what the hospital billed for care, and they are generally an overestimate of the actual

expenses incurred in the production of services. To convert charges into cost estimates, we used the publically available HCUP Cost-to-Charge Ratio File for 2012 [20]. This file contains hospital-specific cost-to-charge ratios for all-payer costs for every hospital in the 2012 NIS. The cost-to-charge ratios are based on cost information from all-payer hospital accounting reports collected by the Centers for Medicare and Medicaid Services.

Independent variables

All diagnosis variables were dichotomous, based on principal and secondary ICD-9-CM codes listed in discharge records. Hospitalizations were categorized into two groups: an ALD group for stays that included any principal or secondary diagnostic codes for alcoholic fatty liver, acute alcoholic hepatitis, alcoholic cirrhosis or unspecified alcoholic liver damage; and a non-ALD group for stays that included other alcohol-related diagnosis that are severe and often fatal (e.g. alcohol withdrawal, alcohol-induced psychotic disorders, alcoholic cardiomyopathy). Similarly, variables were created to categorize hospitalizations by whether or not they included diagnoses of acute kidney disease, ascites, coronary atherosclerosis, essential hypertension, hepatic coma, hepatocellular carcinoma, lipid disorders, protein-calorie malnutrition, portal hypertension, upper GI bleeding and viral hepatitis. A list of the 149 different ICD-9-CM codes used to create these variables is available in Supporting information, Appendix S1.

The analysis included dichotomous variables representing procedures that are often provided to patients hospitalized with ALD [19,21], based on the Clinical Classifications Software (CCS) for Services and Procedures, a method for classifying in-patient ICD-9-CM procedure codes into clinically meaningful categories [22]. The procedure categories included were abdominal paracentesis, enteral and parenteral nutrition, liver biopsy, psychological/psychiatric evaluation and treatment, renal replacement therapy (i.e. dialysis), respiratory intubation/mechanical ventilation and upper GI endoscopy/biopsy.

Patient demographic variables included in the analysis were age (continuous), sex (female versus male) and race/ethnicity (black, Hispanic, non-Hispanic white and 'other'—the last of which included groups such as Asian, Pacific Islander and American Indian). Expected primary payer was included as a five-category variable for Medicare, Medicaid, private insurance, uninsured (including self-pay and no charge) and 'other' (including Worker's Compensation and other government programs). Although patient-level income information is not available in the NIS, the data set includes a four-category Zip-code-level income variable that provides a quartile classification of the estimated median household income of residents in the patient's Zip code.

Data analysis

For the two groups of patients with and without ALD, we compared summary statistics on total costs (e.g. weighted average, median), as well as patient demographics, diagnoses and procedure categories (e.g. weighted percentages and frequencies). The Stata software package 'svy: tab' command and 'count' option were used to estimate weighted population frequencies and percentages from the sample, while accounting for stratification and clustering [23]. We compared average total costs across categories of patient race/ethnicity, sex, payer and diagnoses (with and without ascites, hepatic coma, etc.). We then estimated the marginal in-patient cost of ALD—i.e. the difference in average total costs for hospitalizations involving ALD versus those involving other alcohol-related diagnoses, controlling for other factors. The marginal costs of ALD, other diagnoses, procedures and patient demographics were estimated using a multivariable generalized linear model with a gamma distribution and a log link to accommodate skewed cost data.

The Oaxaca-Blinder regression decomposition technique was used to examine potential mechanisms for the difference in costs between patients with ALD and patients with other alcohol-related diagnoses. Regression decomposition allowed us to estimate how much of the total difference in costs between these two patient groups could be accounted for by each of the independent variables, while holding constant the other independent variables [24]. The decomposition method partitions the difference in estimated costs between the two groups into two components: the 'explained' component, which is due to the fact that the groups have, on average, different values for the known characteristics (the variables included in the model); and the 'unexplained' component, which reflects the differential effects that the characteristics have on each group, as well as other unobserved characteristics. In the results, we focus on the explained component; that is, the degree to which the different distributions of diagnostic, demographic and procedure characteristics between the ALD and non-ALD groups account for the difference in hospital costs. The approach is referred to as a 'decomposition', because the total difference in the outcome is broken down into the percentage of difference attributable to each independent variable (in contrast to procedures that would generate a single statistic for the entire set of variables in the model, such as the R^2 or C statistics). Regression decomposition has been used extensively by labor economists to account for factors driving gender disparities in income [25], with more recent applications in health focusing on disparities in service use by race/ethnicity [26] and hepatitis C prevalence by military veteran status [27].

All analyses were conducted with the Stata version 14.0 software program to adjust for the NIS sampling

design and produce weighted national estimates. The Stata 'margin' command [23] was used to generate predictive margins (i.e. marginal cost estimates) in the multivariable model, and the 'oaxaca' command [28] was used to perform the regression decomposition. As a model specification test we used the 'linktest' command in Stata, which is based on the idea that if a regression is specified fully, one should not find any additional variables that are significant except by chance. The procedure creates two new variables, the variable of prediction ('_hat') and the variable of squared prediction ('_hatsq'), and then refits the model using these two variables as predictors. The variable '_hat' should be significant, because it is the predicted value. By contrast, '_hatsq' should not be significant, because if the model is specified fully, the squared predictions should not have much explanatory power.

RESULTS

Alcoholic liver disease accounted for 20% of all hospitalizations involving alcohol use disorders in US community hospitals in 2012 (Table 1). At \$4.8 billion, these stays represented approximately 25% of aggregate in-patient costs for all stays involving alcohol use disorders (\$19.4 billion). Average costs per stay for patients with ALD were approximately 30% higher than for patients with other alcohol diagnoses (\$13 543 versus \$10 355, $P < 0.001$), and median costs were 37% higher (\$7918 versus \$5777, $P < 0.001$).

Demographic characteristics of patients with ALD and patients with other alcohol-related diagnoses are presented in Table 1. Although the average age of ALD patients was higher than that of non-ALD patients, the percentage of female patients in the two groups was equal. Compared with non-ALD patients, the ALD group had a lower percentage of black patients and a higher percentage of Hispanic patients. Insurance coverage for the two groups was similar, although a lower percentage of ALD patients were uninsured.

Compared with patients with other alcohol-related diagnoses, patients with ALD had more than twice the prevalence of malnutrition and viral hepatitis, more than four times the prevalence of upper GI bleeding; approximately 10 times the prevalence of liver cancer; 25 times the prevalence of ascites and hepatic coma; and more than 35 times the prevalence of portal hypertension (Table 1).

Table 1 shows that a greater percentage of patients with ALD received five of the six procedures included in the analysis compared with non-ALD patients (Table 1). Specifically, a greater percentage of ALD patients than non-ALD patients received abdominal paracentesis, enteral/parenteral nutrition, liver biopsy, respiratory intubation/mechanical ventilation and upper GI endoscopy/biopsy. By contrast, a lower percentage of ALD

than non-ALD patients received psychological/psychiatric care.

As shown in Table 2, average unadjusted costs of ALD stays were \$3188 higher than those for stays involving other alcohol-related diagnoses. Clinical sequelae and comorbidities were also associated with higher costs. Compared with privately insured stays, unadjusted average costs were higher for Medicare and lower for the uninsured. Table 2 also shows that unadjusted average costs were higher for Hispanics and patients of 'other' race/ethnicity than for whites (Table 2). Unadjusted average costs for black patients were slightly lower than those for whites.

Regression and decomposition analyses

After multivariable adjustments, the difference in average costs for hospitalizations involving ALD and stays involving other alcohol-related diagnoses was not significant (Table 3). Significant marginal costs of diagnoses associated with ALD, however, ranged from \$322 for viral hepatitis to \$6501 for protein-calorie malnutrition. Among the procedures, marginal costs ranged from \$1639 for abdominal paracentesis to \$17 767 for respiratory intubation and mechanical ventilation. The decomposition analysis shows that the diagnoses included in the model accounted for approximately 54% of the cost difference between ALD and non-ALD patients, with the greatest percentages attributable to acute kidney disease (15.6%), protein-calorie malnutrition (11.7%) and hepatic coma (9.8%). Differences in the use of procedures accounted for approximately 34% of the cost difference between ALD and non-ALD patients, with the greatest percentage attributable to respiratory intubation and mechanical ventilation (8.6%), upper GI endoscopy or biopsy (8.5%) and abdominal paracentesis (6.2%).

After multivariable adjustments, the cost for Medicare hospitalizations was slightly lower than costs of the privately insured. Average costs of hospitalizations for uninsured patients continued to be lower than for the privately insured. In the decomposition analysis, the categories for insurance coverage accounted for 0.6% of the difference in costs between ALD and non-ALD hospital stays. The marginal costs of stays were lower for female patients compared with male patients, but higher for Hispanics and patients of 'other' racial/ethnic backgrounds compared with whites. The decomposition showed that sex and race/ethnicity accounted for 0.0 and 2.5%, respectively, of the cost difference for ALD and non-ALD patients, whereas age accounted for 7.6% of the difference. Taken together, differences in demographic characteristics between the two groups accounted for approximately 11% of the higher average cost of hospitalizations involving ALD, compared with other stays. The results from the

Table 1 Characteristics of hospital stays of adult patients with alcoholic liver disease and patients with other alcohol-related diagnoses; United States, 2012.

	<i>Alcoholic liver disease</i>	<i>Other alcohol-related diagnoses^a</i>
Unweighted <i>n</i>	72 531	287 047
Weighted <i>n</i>	362 665	1 435 236
Weighted %	20.2	79.8
In-patient costs, \$US		
Aggregate (billions)	4.8	14.6
Mean (SE)	13 543 (173)	10 355 (121)***
Median (IQR)	7918 (9628)	5777 (7639)***
Range	1413–889 132	874–864 207
Demographics		
Female, % (<i>n</i>) ^b	27.9 (20 222)	27.9 (80 201)
Age, mean (SE)	53.7 (0.07)	50.2 (0.10)***
Race/ethnicity, % (<i>n</i>) ^b		
Black	10.7 (7721)	17.4 (49 806)***
Hispanic	14.1 (10 256)	8.6 (24 654)
Other	5.8 (4199)	4.8 (13 885)
White	64.6 (46 864)	63.4 (182 097)
Missing race/ethnicity	4.8 (3490)	5.8 (16 597)
Expected primary payer, % (<i>n</i>) ^b		
Medicare	29.9 (21 584)	28.3 (81 090)***
Medicaid	25.6 (18 499)	23.5 (67 119)
Other	6.0 (4368)	6.3 (18 121)
Uninsured	16.5 (11 921)	18.1 (51 907)
Private	22.0 (15 938)	23.8 (68 018)
Community-level income quartiles		
1st quartile (lowest income)	32.2 (23 397)	33.7 (96 664)***
2nd quartile	24.1 (17 506)	23.5 (67 418)
3rd quartile	22.2 (16 087)	21.1 (60 670)
4th quartile (highest income)	18.2 (13 168)	17.4 (49 826)
Missing income	3.3 (2357)	4.3 (12 395)
Diagnoses, % (<i>n</i>) ^b		
Ascites	28.4 (20 602)	1.1 (3284)***
Coronary atherosclerosis	9.8 (7080)	12.7 (36 306)***
Essential hypertension	35.9 (26 033)	40.3 (115 599)***
Hepatic coma	22.7 (16 480)	0.9 (2518)***
Kidney disease	19.3 (13 981)	8.3 (23 770)
Lipid disorders	11.8 (8534)	18.8 (53 883)
Liver cancer	2.9 (2088)	0.3 (705)***
Portal hypertension	22.2 (16 083)	0.6 (1839)***
Protein–calorie malnutrition	12.3 (8946)	4.6 (13 263)***
Upper GI bleeding	11.5 (8323)	2.4 (6770)***
Viral hepatitis	16.4 (11 859)	6.9 (19 654)***
Procedure categories, % (<i>n</i>) ^b		
Abdominal paracentesis	17.4 (12 587)	0.7 (1905)***
Enteral and parenteral nutrition	2.1 (1505)	1.3 (3838)***
Liver biopsy	1.2 (903)	0.2 (667)***
Psychological/psychiatric services	0.4 (270)	2.5 (7095)***
Renal replacement therapy	3.0 (2180)	1.2 (3030)***
Respiratory intubation and mechanical ventilation	9.0 (6543)	7.0 (19 998)***
Upper GI endoscopy, biopsy	13.8 (9973)	4.3 (12 400)***

GI = gastrointestinal; IQR = interquartile range; SE = standard error; ^aThe 'other alcohol-related diagnoses' hospital stays included those with the following ICD-9-CM-based diagnosis codes: acute alcohol intoxication, alcohol abuse, alcoholic beverages poisoning, alcoholic cardiomyopathy, alcoholic gastritis, alcoholic polyneuropathy, alcohol-induced persisting amnesic disorder, alcohol-induced persisting dementia, alcohol-induced psychotic disorder with delusions, alcohol-induced psychotic disorder with hallucinations, alcohol-induced sleep disorders, other alcohol-induced disorders, unspecified alcohol-induced mental disorders, other and unspecified alcohol dependence, alcohol withdrawal, alcohol withdrawal delirium, idiosyncratic alcohol intoxication and other specified alcohol-induced mental disorders. ^bPercentages are weighted, frequencies are unweighted. ****P* < 0.001.

Table 2 Bivariate analyses of in-patient costs by select diagnoses and patient demographics; hospital stays of adult patients with alcohol-related diagnoses; United States, 2012.

<i>Diagnoses</i>	<i>Was diagnosis present?</i>	<i>Unadjusted average costs</i>	<i>Difference in average costs (95% CI)</i>
Alcoholic liver disease	Yes	13 543.4	3188.4 (2917.8, 3459.0)***
	No	10 355.0	—
Ascites	Yes	17 646.4	7120.8 (6534.2, 7707.4)***
	No	10 525.6	—
Coronary atherosclerosis	Yes	14 010.7	3423.7 (3176.1, 3671.3)***
	No	10 587.0	—
Essential hypertension	Yes	11 204.1	340.4 (201.1, 479.7)***
	No	10 863.7	—
Hepatic coma	Yes	18 384.6	7798.8 (7172.3, 8425.4)***
	No	10 585.8	—
Kidney disease	Yes	21 376.6	11 587.0 (11 017.5, 12 156.6)***
	No	9789.6	—
Lipid disorders	Yes	12 117.1	1350.9 (1172.6, 1529.3)***
	No	10 766.2	—
Liver cancer	Yes	17 066.2	6116.1 (4883.0, 7349.2)***
	No	10 950.1	—
Portal hypertension	Yes	16 421.6	5706.9 (5140.1, 6273.7)***
	No	10 714.7	—
Protein-calorie malnutrition	Yes	24 770.3	14 660.7 (13 889.7, 15 431.7)***
	No	10 109.6	—
Upper GI bleeding	Yes	15 860.7	5075.6 (4667.8, 5483.4)***
	No	10 785.1	—
Viral hepatitis	Yes	11 459.6	506.6 (199.5, 813.6)***
	No	10 953.0	—
<i>Demographics</i>			
Expected primary payer			
Medicare		12 237.5	1216.6 (988.7, 1444.5)***
Medicaid		11 054.2	33.4 (−315.6, 382.3)
Other		10 978.7	−42.2 (−579.6, 495.3)
Uninsured		8940.6	−2080.3 (−2435.7, −1724.9)***
Private ^a		11 020.9	—
Patient race/ethnicity			
African American/black		10 229.2	−681.6 (−1023.6, −339.6)***
Hispanic		12 054.7	1143.9 (236.9, 2050.9)*
Other		12 413.8	1503.0 (639.3, 2366.6)***
Missing race		11 068.7	157.9 (−575.8, 891.7)
White ^a		10 910.8	—
Patient sex			
Female		9935.8	−1472.9 (−1628.4, −1317.5)***
Male ^a		11 408.8	—

^aReference category in calculating and testing differences in average costs. * $P < 0.05$; *** $P < 0.001$. CI = confidence interval; GI = gastrointestinal.

'linktest' model specification test (described in the Methods) showed that the variable of squared prediction ('_hatsq') was statistically significant ($P < 0.0001$), suggesting that relevant variables may have been omitted from the model.

DISCUSSION

This analysis estimated that average costs per hospital stay for patients with ALD were approximately 30% higher than those for patients with other alcohol-related diagnoses. Compared with publically available national

benchmarks for other chronic conditions [29], average costs for ALD stays as estimated in this analysis (\$13 543) were also higher than those for stays with a principal diagnosis of diabetes mellitus with complications (\$9361), congestive heart failure (\$11 412) and hypertension with complications (\$11 673), and comparable to stays for cancers of the breast (\$13 051) and uterus (\$13 859). To our knowledge, this study is the first to quantify the impact of the more common sequelae and comorbidities of ALD on total in-patient costs. Results showed that differences between the two groups in the prevalence

Table 3 Multivariable analysis and decomposition of the difference in total in-patient costs between hospital stays involving alcoholic liver disease and stays involving other alcohol-related diagnoses; United States, 2012.

	Multivariable analysis		Decomposition ^a
	PM	95% CI	% of ALD/non-ALD cost difference ^b
<i>Diagnoses (reference)</i>			
Alcoholic liver disease (none)	−109.9	(−278.9, 59.0)	—
Ascites (none)	901.9	(616.0, 1187.7)***	7.1%
Coronary atherosclerosis (none)	2115.2	(1934.9, 2295.4)***	−1.1%
Essential hypertension (none)	272.6	(170.8, 374.3)***	−0.4%
Hepatic coma (none)	1064.7	(840.7, 1288.7)***	9.8%
Kidney disease (none)	3817.1	(3582.0, 4052.1)***	15.6%
Lipid disorders (none)	599.8	(463.0, 736.5)***	−0.2%
Liver cancer (none)	1250.8	(517.7, 1984.0)	1.4%
Portal hypertension (none)	752.7	(504.1, 1001.3)***	6.2%
Protein-calorie malnutrition (none)	6501.0	(5956.5, 7045.3)***	11.7%
Upper GI bleeding (none)	207.1	(0.63, 413.6)*	2.5%
Viral hepatitis (none)	322.0	(101.7, 542.3)**	1.4%
Total ALD/non-ALD cost difference explained by diagnosis variables ^c			54.0%
<i>Demographic characteristics (reference)</i>			
Female (male)	−891.0	(−1007.3, −774.7)***	0.0%
Age	99.0	(93.2, 104.9)***	7.6%
Race/ethnicity (white)			−0.3% ^d
African American/black	52.1	(−260.4, 364.5)	1.5%
Hispanic	1494.9	(629.2, 2360.6)***	1.1%
Other	1467.4	(775.6, 2159.3)***	0.2%
Missing race	771.1	(81.4, 1460.8)*	0.0%
Expected primary payer (private)			−0.1% ^e
Medicare	−535.6	(−727.1, −344.2)***	−0.2%
Medicaid	177.8	(−87.7, 443.3)	0.5%
Other	642.0	(152.7, 1131.2)**	0.0%
Uninsured	1056.4	(−1400.6, −712.2)***	0.4%
Community-level income (1st quartile, lowest income)			0.6% ^f
2nd quartile	310.2	(110.5, 509.8)**	−0.2%
3rd quartile	946.3	(695.7, 1197.0)***	0.1%
4th quartile (highest income)	1761.5	(1380.3, 2142.8)***	0.3%
Missing income	408.6	(−115.7, 933.0)	−0.2%
Total ALD/non-ALD cost difference explained by demographic variables ^c			11.3%
<i>Procedure categories (reference)</i>			
Abdominal paracentesis (none)	1639.4	(1311.8, 1967.1)***	6.2%
Enteral and parenteral nutrition (none)	13 039.0	(12 228.2, 13 849.7)***	2.3%
Liver biopsy (none)	11 163.9	(9238.9, 13 088.8)***	3.1%
Psychological/psychiatric services (none)	1793.8	(511.9, 3075.6)**	0.0%
Renal replacement therapy (none)	6680.4	(6079.0, 7281.9)***	4.8%
Respiratory intubation and mechanical ventilation (none)	17 767.1	(17 114.2, 18 419.9)***	8.6%
Upper GI endoscopy, biopsy (none)	2732.6	(2524.2, 2941.0)***	8.5%
Total ALD/non-ALD cost difference explained by procedure variables ^c			33.5%
Total ALD/non-ALD cost difference explained ^a	—		98.8%

ALD = alcoholic liver disease; CI = confidence interval; GI = gastrointestinal; non-ALD = alcohol-related diagnoses other than those affecting the liver; PM = predictive margin; SE = standard error. ^aDecomposition results are reported as the percentage of the difference in average costs of stays with and without ALD that is attributable to each independent variable. Together, the variables explain 99.0% of the difference in costs between the two groups. ^bThe non-ALD (other alcohol-related diagnoses) hospital stays included those with the following ICD-9-CM-based diagnosis codes: acute alcohol intoxication, alcohol abuse, alcoholic beverages poisoning, alcoholic cardiomyopathy, alcoholic gastritis, alcoholic polyneuropathy, alcohol-induced persisting amnesic disorder, alcohol-induced persisting dementia, alcohol-induced psychotic disorder with delusions, alcohol-induced psychotic disorder with hallucinations, alcohol-induced sleep disorders, other alcohol-induced disorders, unspecified alcohol-induced mental disorders, other and unspecified alcohol dependence, alcohol withdrawal, alcohol withdrawal delirium, idiosyncratic alcohol intoxication and other specified alcohol-induced mental disorders. ^cThis row is sum of the percentages for this subgroup of variables. ^dPercentage of difference attributable to reference group (white). ^ePercentage of difference attributable to reference group (private insurance). ^fPercentage of difference attributable to reference group (lowest community income quartile). * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

of ascites, malnutrition, upper GI bleeding and other conditions together accounted for 54% of the higher average cost of hospitalizations involving ALD, compared with other stays. Identifying the cost drivers of ALD hospital care is important, as hospitals face increasing pressure to contain costs, taking on greater financial risk for the care of defined populations [30]. These findings suggest some of the savings for payers that could result from policies and programs to prevent liver injury and its consequences among patients who misuse alcohol.

Of the diagnoses included in the analysis, protein-calorie malnutrition had the highest marginal cost. People with alcohol use disorders develop malnutrition because of inadequate nutrient intake and an impaired ability to metabolize nutrients, often as a result of liver injury. Protein-calorie malnutrition was more than twice as common among ALD patients than those with other alcohol-related diagnoses, and accounted for approximately 12% of the higher costs of ALD-related stays. Previous work has shown that some ALD treatments (e.g. anabolic steroids) are less effective among patients with more severe forms of malnutrition [14]. The current finding may, in part, reflect the cost of trying additional therapies in severely malnourished patients with ALD for whom initial treatment options failed, as well as greater disease severity among patients diagnosed with malnutrition [12].

Of the procedure categories included in this analysis, respiratory intubation and mechanical ventilation had the highest marginal costs. Mechanical ventilation is often required for patients experiencing alcohol-withdrawal seizures. Respiratory infections are common among individuals who abuse alcohol and tobacco, and in-patient treatment of pneumonia in these patients often involves intubation and mechanical ventilation. The finding that these services accounted for approximately 9% of the increased cost of ALD stays probably reflects the immunosuppression and respiratory infections seen in the ALD patient population [31].

The average cost of hospitalizations for women with any alcohol-related diagnosis was lower than that for men, which is consistent with the literature showing that men generally develop more severe alcohol use disorders. At the same time, numerous studies have shown that women have greater susceptibility to ALD than do men, believed to be due in part to sex differences in body fat and alcohol dehydrogenase [9]. Although women may be more susceptible to ALD, the decomposition suggests that among patients whose disease has advanced far enough to require hospitalization, patient sex has virtually no implications for ALD in-patient costs. In the current study, Hispanic ethnicity was associated with higher marginal costs of care, which probably reflects greater disease severity and comorbidity. Previous work has shown that mortality due to alcoholic cirrhosis is higher in Hispanic than white men [32].

Patients of 'other' backgrounds also had higher marginal costs than non-Hispanic whites, perhaps because this category included groups (e.g. Asians) that generally produce less alcohol dehydrogenase [13], an enzyme needed to break down alcohol in the stomach, creating greater vulnerability to alcoholic liver injury, which would, in turn, drive increased costs. The decomposition finding that age accounted for such a large percentage of the marginal cost of ALD (7.6%) is expected, as age is a strong predictor of poor prognosis in ALD [33,34].

Limitations

This study had several limitations. Although the sampling frame covered more than 95% of the US population in 2012, the NIS does not include the Veterans Health Administration and other federal hospitals, or hospitals where the average length of stay is 30 days or longer. Further, these estimates do not include non-hospital spending on ALD; however, hospitals continue to constitute a large proportion of health-care spending, making an analysis of this type of care useful for stakeholders concerned with costs. Because of the cross-sectional study design, the relationships between costs, diagnoses, and procedures that we estimated may be associational and not causal. The set of ICD-9-CM codes used to define the ALD variable is broad, including alcoholic fatty liver disease, which is highly prevalent among people who abuse alcohol, but may not have in-patient costs associated with it; this may over- or underestimate some of the results presented. Relevant clinical detail is often not available with ICD-9-CM code data, and there is also a possibility of under- or over-reporting of diagnoses and procedures within hospitals. Previous work on liver disease hospitalizations, however, underscores some of the strengths of NIS data in this area. In particular, in a study of hepatocellular carcinoma (HCC), Kim and colleagues [35] assessed the validity of the 1998 NIS data by comparing the number of HCC-related in-patient deaths in the NIS with that of another, independent data source, the Multiple Cause of Death files from the National Center for Health Statistics. The number of HCC-related deaths reported in the two databases for 1998 were similar, at 3267 and 3171, respectively.

One limitation of administrative data is that they are intended typically for financial and administrative purposes, and they may lack information such as results from laboratory tests and other potentially important clinical or psychosocial information available in electronic health records. The unavailability of these kinds of clinical data in the HCUP NIS database may be a reason why the model specification test suggested that the model did not include all relevant variables (because the parameter estimate for

the variable of squared prediction ('_hatsq') was statistically significant). To the degree that in-patient costs of ALD are driven by other factors we were not able to include in our model, the potential for omitted variables bias needs to be acknowledged. Although the NIS is the most comprehensive, nationally representative source of data on US hospital care, future work exploring potential linkages between this database and electronic health record data in ways that do not compromise patient privacy and confidentiality could, conceivably, improve model specification by allowing for the inclusion of variables with richer clinical information.

CONCLUSION

The fact that ALD is both costly and preventable raises questions about how public health and medical interventions can reduce the incidence of this condition. The current analysis showed that the costs of hospital care for patients with ALD were higher than those for patients with other alcohol-related diagnoses. These increased costs are driven by specific clinical sequelae and comorbidities, with protein-calorie malnutrition accounting for a substantial portion of the difference. Malnutrition is a preventable condition, making it a potentially useful target in efforts to reduce the financial burden of ALD and other alcohol-related conditions on public and private payers. To our knowledge, there is no specific evidence base for interventions to prevent liver damage and its associated treatment costs among people who abuse alcohol. However, an estimated 90% of people who abuse alcohol develop some degree of alcoholic fatty liver [36,37]. Both fatty liver and alcoholic hepatitis are largely reversible, which suggests that more general public health and medical interventions to prevent or treat alcohol abuse could conceivably prevent the progression of these conditions to alcoholic cirrhosis. Given the increasing consumption of alcohol in the United States [1,2], the role of such interventions in preventing ALD will be important for hospitals responsible for the care of patients with alcohol use disorders.

Declaration of interests

None.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Appendix S1 ICD-9-CM diagnosis codes defining comorbidities and sequelae of alcoholic liver disease.